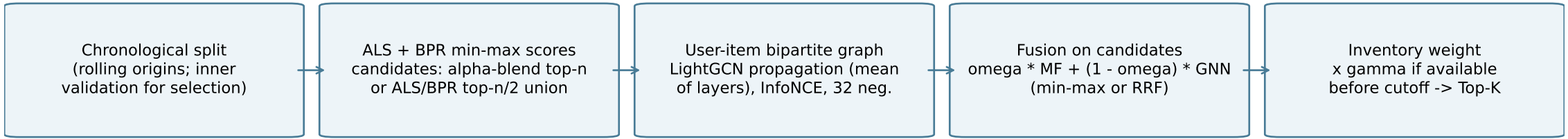

Pair weight w = view 1 / cart 6 / purchase 12, decayed with a 7-day half-life: ALS confidence 1 + alpha * w;
BPR loss weight w (popularity negatives); GNN InfoNCE loss weight w (unweighted propagation, edge dropout 0.1, uniform negatives).

No Isolation Forest in the final model. All parameters chosen on inner validation; final models refit with fixed epochs on the full pre-test window.